Monitoring Docker Infrastructure

Elsewhere in this issue, the installation of Docker has been covered. This article deals with how Docker infrastructure can be monitored.

Docker brings in portability to technology-intensive software development environments. As a result, it has become quite popular among developers to demonstrate a proof-of-concept, technology preview release through the use of Docker containers.

At the same time, Docker runs UNIX processes on Linux with strong guarantees of isolation, thanks to Linux kernel technologies like namespaces and control groups (cgroups). This has disrupted the entire deployment and operations space and proven Docker to be a great building block for automating software infrastructure, namely, large-scale Web deployments, databases, private Platforms as a Service (PaaS), continuous deployment systems and service-oriented infrastructures (SOA), among others.

In simple terms, the Docker runtime itself is a container based PaaS platform.

Blueprints of monitoring infrastructure

In any infrastructure, the resources that are at stake are computing power (CPU), memory (RAM), networking (LAN) and storage (HDD). Virtualisation brought in hardware resource utilisation and configurable limits, leading to efficiency. However, containerisation disrupted the entire game with frugal system resource utilisation, powered by operating system software capabilities, thanks to namespaces and cgroups.

As a result, one can run thousands of 'dockerised' portable apps on commodity hardware, driving forward significant efficiency and scale on the same platform that earlier struggled under the virtualisation workload of tens of operating systems.

Given this flexibility, it is entirely possible that a bunch of resource-hungry applications can significantly reduce the performance of the other 'dockerised' apps. The objective of monitoring is to precisely address this important case by collecting operating system level metrics and having a mechanism to manage the apps to ensure scalability. This article highlights monitoring within this context.

Namespaces in the Linux kernel

In the Linux kernel, the six namespaces are *mount, uts, ipc, pid, net* and *user,* which provide isolation of the system's resources. It is important to understand that namespaces and control groups are orthogonal by design.